

Sri Lanka Institute of Information Technology



SE3060 - Database Systems Laboratory Worksheet 3

Student ID	Student Name	Email	Contact Number
IT23168404	K.S. Shiwanthi Dias	it23168404@my.sliit.lk	0760369961

1. Create the required types and object tables in your database and insert the given sample data.

Create Types

```
CREATE TYPE address_t AS OBJECT (  
    street_no  NUMBER,  
    street_name VARCHAR2(50),  
    suburb    VARCHAR2(50),  
    state     VARCHAR2(20),  
    pin       VARCHAR2(10)  
)  
/
```

```
CREATE TYPE investment_t AS OBJECT (  
    company   VARCHAR2(30),  
    price     NUMBER(8,2),  
    purchase_dt DATE,  
    qty       NUMBER  
)  
/
```

```
CREATE TYPE investment_table_t AS TABLE OF investment_t  
/
```

```
CREATE TYPE client_t AS OBJECT (  
    fname     VARCHAR2(30),  
    lname     VARCHAR2(30),  
    address   address_t,  
    investments investment_table_t  
) NESTED TABLE investments STORE AS investments_nt;  
/
```

```
CREATE TYPE exchange_varray_t AS VARRAY(5) OF VARCHAR2(30);  
/
```

```

CREATE TYPE stock_t AS OBJECT (
    company    VARCHAR2(30),
    curr_price  NUMBER(8,2),
    exchanges   exchange_varray_t,
    last_dividend NUMBER(8,2),
    eps        NUMBER(8,2)
)
/

```

Create tables

```

CREATE TABLE clients_tab OF client_t
(PRIMARY KEY (fname, lname))
NESTED TABLE investments STORE AS investments_store;

```

```

CREATE TABLE stocks_tab OF stock_t
(PRIMARY KEY (company));

```

Insert data into the client table

```

INSERT INTO clients_tab VALUES (
    client_t(
        'John', 'Smith',
        address_t(3, 'East Av', 'Bentley', 'WA', '6102'),
        investment_table_t(
            investment_t('BHP', 12.00, DATE '2001-10-02', 1000),
            investment_t('BHP', 10.50, DATE '2002-06-08', 2000),
            investment_t('IBM', 58.00, DATE '2000-02-12', 500),
            investment_t('IBM', 65.00, DATE '2001-04-10', 1200),
            investment_t('INFOSYS', 64.00, DATE '2001-08-11', 1000)
        )
    )
)
/

```

```

INSERT INTO clients_tab VALUES (
  client_t(
    'Jill', 'Brody',
    address_t(42, 'Bent St', 'Perth', 'WA', '6001'),
    investment_table_t(
      investment_t('INTEL', 35.00, DATE '2000-01-30', 300),
      investment_t('INTEL', 54.00, DATE '2001-01-30', 400),
      investment_t('INTEL', 60.00, DATE '2001-10-02', 200),
      investment_t('FORD', 40.00, DATE '1999-10-05', 300),
      investment_t('GM', 55.50, DATE '2000-12-12', 500)
    )
  )
);

```

Insert date into the stock table

```

INSERT INTO stocks_tab VALUES (
  stock_t('BHP', 10.50, exchange_varray_t('Sydney','New York'), 1.50, 3.20)
);

```

```

INSERT INTO stocks_tab VALUES (
  stock_t('IBM', 70.00, exchange_varray_t('New York','London','Tokyo'), 4.25, 10.00)
);

```

```

INSERT INTO stocks_tab VALUES (
  stock_t('INTEL', 76.50, exchange_varray_t('New York','London'), 5.00, 12.40)
);

```

```

INSERT INTO stocks_tab VALUES (
  stock_t('FORD', 40.00, exchange_varray_t('New York'), 2.00, 8.50)
);

```

```

INSERT INTO stocks_tab VALUES (
  stock_t('GM', 60.00, exchange_varray_t('New York'), 2.50, 9.20)
);

```

```

INSERT INTO stocks_tab VALUES (
  stock_t('INFOSYS', 45.00, exchange_varray_t('New York'), 3.00, 7.80)
);

```

2. Answer the queries using the object relational tables

- a) Select c.fname , c.lname AS client_name, i.company, s.curr_price,
s.last_devidend, s.eps
From clients_tab c, table(c.investments) i ,
Stocks_tab s
Where i.company = s.company
- b) Select c.fname, c.lname as client_name, i.company, SUM(i.qty) as total_qty
From clients_tab c, table(c.investments) i
Group by c.fname, c.lname, i.compnay
- c) Select i.company, c.fname, c.lname as client_name, sum(i.qty) as qty_held,
sum(i.qty * s.curr_price) as curr_value
From clinets_tab c, table(c.investments) i, stocks_tab s
Where i.compnay = s.company and 'New York' member of s.exchnages
Group by c.fname, c.lname, i.compnay;
- d) SELECT c.fname , c.lname AS client_name, SUM(i.qty * i.price) AS
total_purchase_value
FROM clients_tab c, TABLE(c.investments) i
GROUP BY c.fname, c.lname;
- e) SELECT c.fname , c.lname AS client_name, SUM(i.qty * s.curr_price)
SUM(i.qty * i.price) AS book_profit
FROM clients_tab c,
TABLE(c.investments) i,
stocks_tab s
WHERE i.company = s.company
GROUP BY c.fname, c.lname;
3. Suppose John sold all his INFOSYS stocks to Jill, and Jill sold all her GM stocks to John today at the current prices. Update the database for these two transactions. Use the query 3(a) to check if the update worked correctly. (Here, the term update does not necessarily mean using the update statement of SQL.)

```
INSERT INTO TABLE(  
    SELECT c.investments  
    FROM clients_tab c  
    WHERE c.fname='Jill' AND c.lname='Brody'  
)  
SELECT *  
FROM TABLE(  
    SELECT c.investments  
    FROM clients_tab c  
    WHERE c.fname='John' AND c.lname='Smith'  
) i  
WHERE i.company='INFOSYS';
```

```
DELETE FROM TABLE(  
    SELECT c.investments  
    FROM clients_tab c  
    WHERE c.fname='John' AND c.lname='Smith'  
) i  
WHERE i.company='INFOSYS';
```

```
INSERT INTO TABLE(  
    SELECT c.investments  
    FROM clients_tab c  
    WHERE c.fname='John' AND c.lname='Smith'  
)  
SELECT *  
FROM TABLE(  
    SELECT c.investments  
    FROM clients_tab c
```

```
WHERE c.fname='Jill' AND c.lname='Brody'  
) i  
WHERE i.company='GM';
```

```
DELETE FROM TABLE(  
  SELECT c.investments  
  FROM clients_tab c  
  WHERE c.fname='Jill' AND c.lname='Brody'  
) i  
WHERE i.company='GM';
```